

IMARTICUS LEARNING

DERIVATIVES – OPTIONS



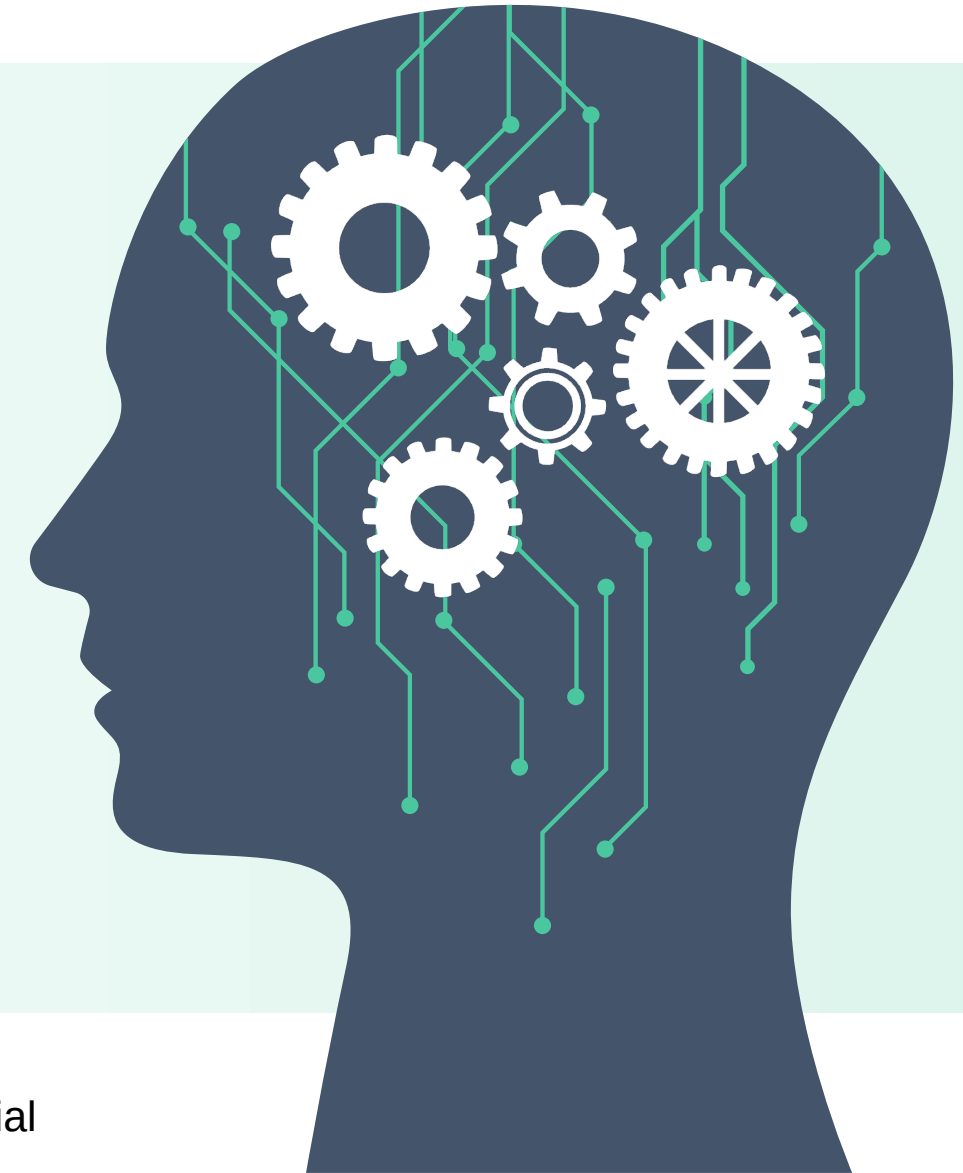
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YOU ARE GOING TO LEARN…

- Introduction to Options
- Features of Options
- Types of Option
- Style of Exercise
- Option Premium & its determinants
- Option Strategies
- ITM/ATM/OTM Status of Options



OPTIONS

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An option contract is a financial derivative that gives the holder the right, but not the obligation, to buy (call option) or sell (put option) a specific underlying asset at a predetermined price (strike price) within a specified period of time (expiration date) and the Option Seller/Writer has an Obligation under the contract . Options are commonly used in financial markets for Hedging/Risk Management, Speculation and Arbitrage.

Features of an option contract:

1. Underlying Asset

An option derives its value from an underlying asset, which can be a stock, bond, commodity, currency, or even another derivative.

2. Type of Option

- **Call Option:** Gives the holder the right to buy the underlying asset at the strike price.
- **Put Option:** Gives the holder the right to sell the underlying asset at the strike price.

OPTIONS

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3. Strike Price

The price at which the underlying asset can be bought (for a call option) or sold (for a put option) if the option is exercised. It is also referred to as the exercise price.

4. Expiration Date

The date when the option contract expires. After this date, the option is no longer valid, and the right to exercise it ceases.

5. Premium

The price the option buyer pays to the option seller for the rights conveyed by the option contract. It represents the cost of holding the option and is determined by various factors, including the current market price of the underlying asset, the strike price, and the time remaining until expiration.

6. Exercise Style

- **American Option:** Can be exercised at any time before or on the expiration date.
- **European Option:** Can only be exercised on the expiration date itself.

OPTIONS

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7. Leverage

Options provide leverage, allowing investors to control a larger position in the underlying asset with a smaller upfront investment. This can amplify both potential gains and losses.

8. Hedging

Options are often used to hedge against adverse price movements in the underlying asset. For example, a stockholder can buy put options to protect against a decline in the stock's value.

9. Speculation

Traders can use options to speculate on the future price movements of the underlying asset, potentially profiting from price changes without owning the asset itself.

10. Risk Limited to Premium

For option buyers, the maximum potential loss is limited to the premium paid for the option. Option sellers, on the other hand, face potentially unlimited losses.

11. Options Chains

Options are typically available with various strike prices and expiration dates, forming an options chain for a particular underlying asset.

12. Settlement Method

Physical Settlement: Involves the actual delivery of the underlying asset if the option is exercised. Cash Settlement: Involves settling the option with a cash payment equal to the difference between the underlying asset's market price and the strike price.

THE BASICS

An option is a financial contract that gives an investor the right, but not the obligation, to either buy or sell an asset at a pre-determined price by a specified date.



Option Buyer

Option Buyer has a Right to exercise the contract or lapse

Option Seller has an Obligation under the contract



Option Seller

Stock price

Underlying Asset

RISK



Stock Option Strike Price

WHAT IS AN OPTION?

Option is a Contract

Right to buy or sell an asset

Predetermined price & quantity

Option Buyer

Sells Rights

Option Seller

Option Contract Characteristics:



Buyer has Right to Choose

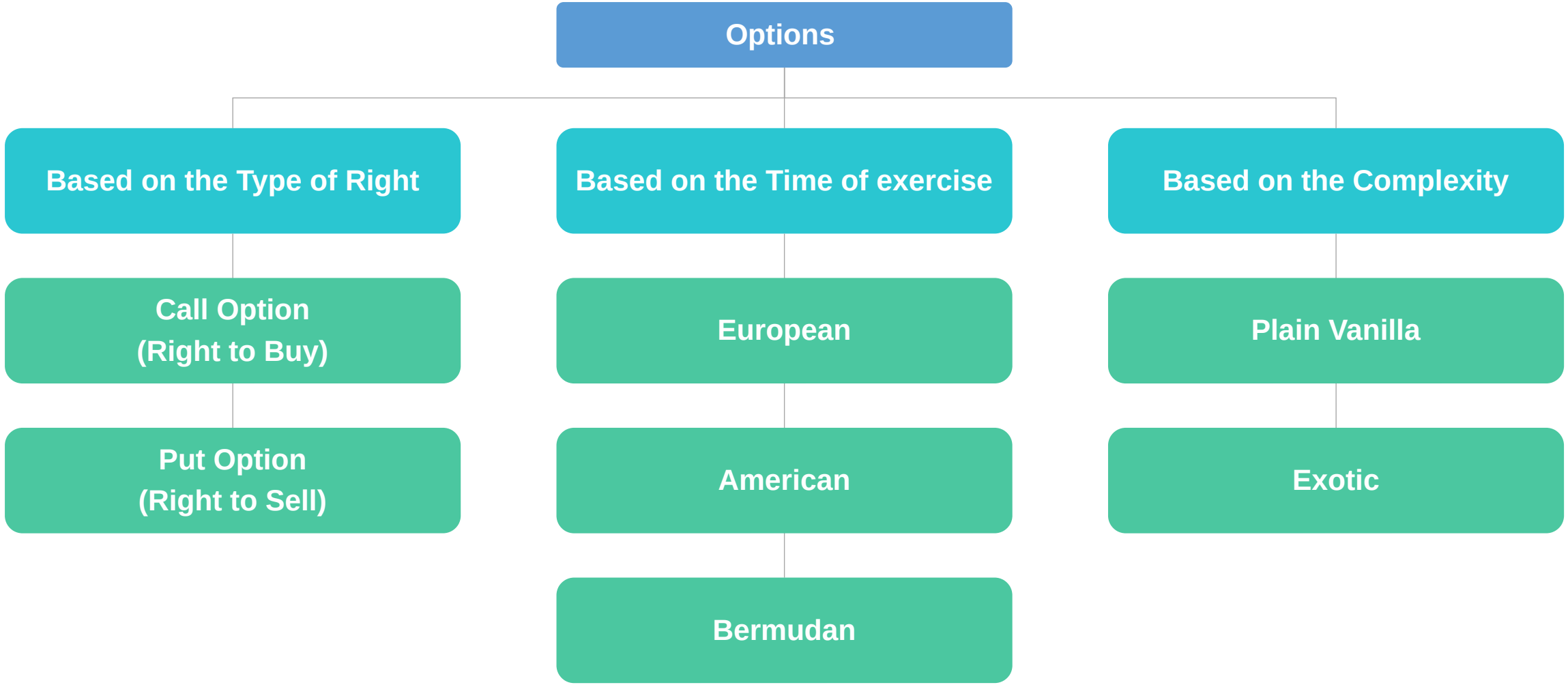


Seller has to follow agreement



Option Buyers Pay Premium Amount

TYPES OF OPTIONS



OPTION STRATEGIES

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Long Call (Buying Call Option)

- Buy Call Option Contract
- Pay a Premium
- Right to BUY the Underlying Asset @ Strike Price
- Holder of the Contract
- Bullish view on the Underlying

Short Call (Selling/Writing Call Option)

- Sell Call Option Contract
- Receives a Premium
- Obligation to SELL the Underlying Asset @ Strike Price
- Writer of the Contract
- Bearish View on the Underelying

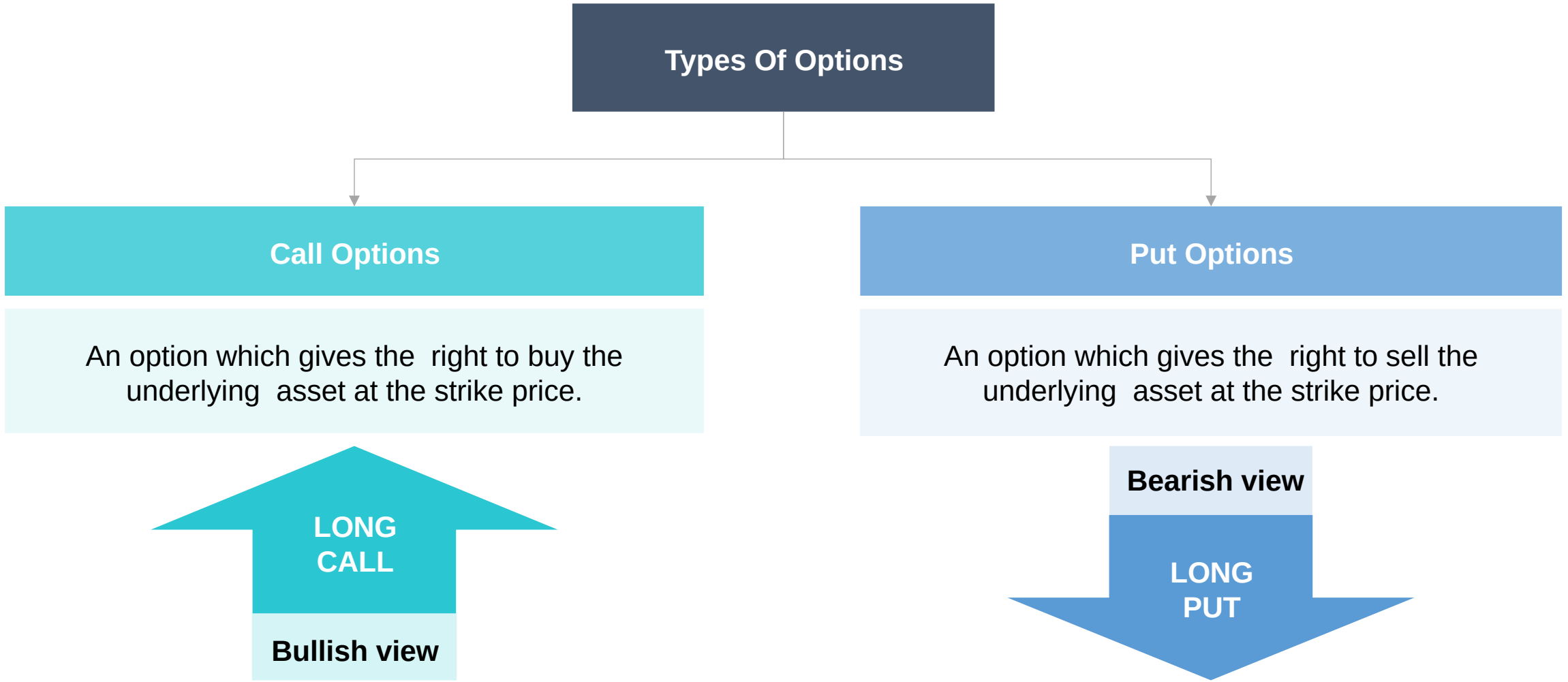
Long Put (Buying Put Option)

- Buy Put Option Contract
- Pay a Premium
- Right to SELL the Underlying Asset @Strike Price
- Holder of the Contract
- Bearish View on the Underelying

Short Put (Selling / Writing Put Option)

- Sell Put Option Contract
- Receives a Premium
- Obligation to BUY the Underlying Asset
- Writer of the Contract
- Bullish view on the Underlying

TYPES OF OPTIONS

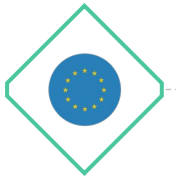


EXERCISE STYLES

Options contracts can have different exercise styles, which determine when the option holder can exercise their rights. The exercise styles for option contracts are:



American Style: With American-style options, the holder (buyer) of the option can exercise their rights at any time before or on the expiration date of the option. This style provides greater flexibility to the option holder, as they can choose the optimal timing for exercising the option based on market conditions.



European Style: European-style options can only be exercised on the expiration date itself. Unlike American-style options, they cannot be exercised prior to the expiration date. This exercise style may limit the flexibility of the option holder but can simplify trading and risk management, as there is no need to monitor or make decisions about early exercise.

It's worth noting that the exercise style is predetermined and specified in the terms of the option contract when it is created. Some options, such as stock options in the United States, are typically American-style, while others, like most index options in the U.S., are European-style. However, the exercise style can vary depending on the specific contract and market. Additionally, some options may have features that allow for early exercise under certain conditions, even if they are primarily European-style options.

EXERCISE STYLES

In addition to **American-style** and **European-style** options, there is a less common exercise style known as "**Bermudan Style**." Bermudan-style options have the following characteristics:

Bermudan Style:

Bermudan-style options can be exercised at specific predetermined dates before or on the expiration date. These predetermined exercise dates are typically specified in the terms of the option contract. While they offer some flexibility compared to European-style options, which can only be exercised on the expiration date, Bermudan-style options are not as flexible as American-style options, which allow for exercise at any time.

Bermudan-style options are named after the island of Bermuda, as they were originally introduced in the 1980s for certain bond options. These options combine elements of both American and European styles, providing exercise opportunities at specific intervals leading up to expiration.

OPTIONS STYLES



American Style Option

Option can be exercised at any time between the date of purchase and the expiration date.

S&P 100



European Style Option

Option can be exercised only on the expiry date.

S&P 500

Vanilla Options

STYLES

Bermudan

May be exercised only on specified dates on or before expiration. These Options are available in OTC Markets



OPTIONS PREMIUM

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An option premium, often simply referred to as the "**premium**," is the price that an option buyer pays to an option seller (or writer) to acquire the rights granted by the option contract. It is a critical component of options trading and has several important features:

1. Cost of the Option

The premium is the cost associated with buying an option. It is an upfront payment made by the option buyer to the seller. The premium's value is determined by various factors, including the current market price of the underlying asset, the option's strike price, the time remaining until expiration, market volatility, and interest rates.

2. Bid-Ask Spread

In the options market, there is typically a bid-ask spread, representing the difference between the price at which buyers are willing to purchase the option (the bid) and the price at which sellers are willing to sell the option (the ask). The premium is usually closer to the ask price for buyers and the bid price for sellers.

OPTIONS PREMIUM

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3. Risk and Reward

For option buyers, the premium represents their maximum potential loss. They can never lose more than the premium paid for the option. For option sellers, the premium represents their maximum potential profit, but their potential losses can be substantial, as they may have to buy or sell the underlying asset at unfavorable prices.

4. Premium Decay

As an option approaches its expiration date, its time value typically decreases more rapidly. This phenomenon is known as time decay or theta decay. It means that, all else being equal, options lose value as they get closer to expiration.

OPTIONS PREMIUM

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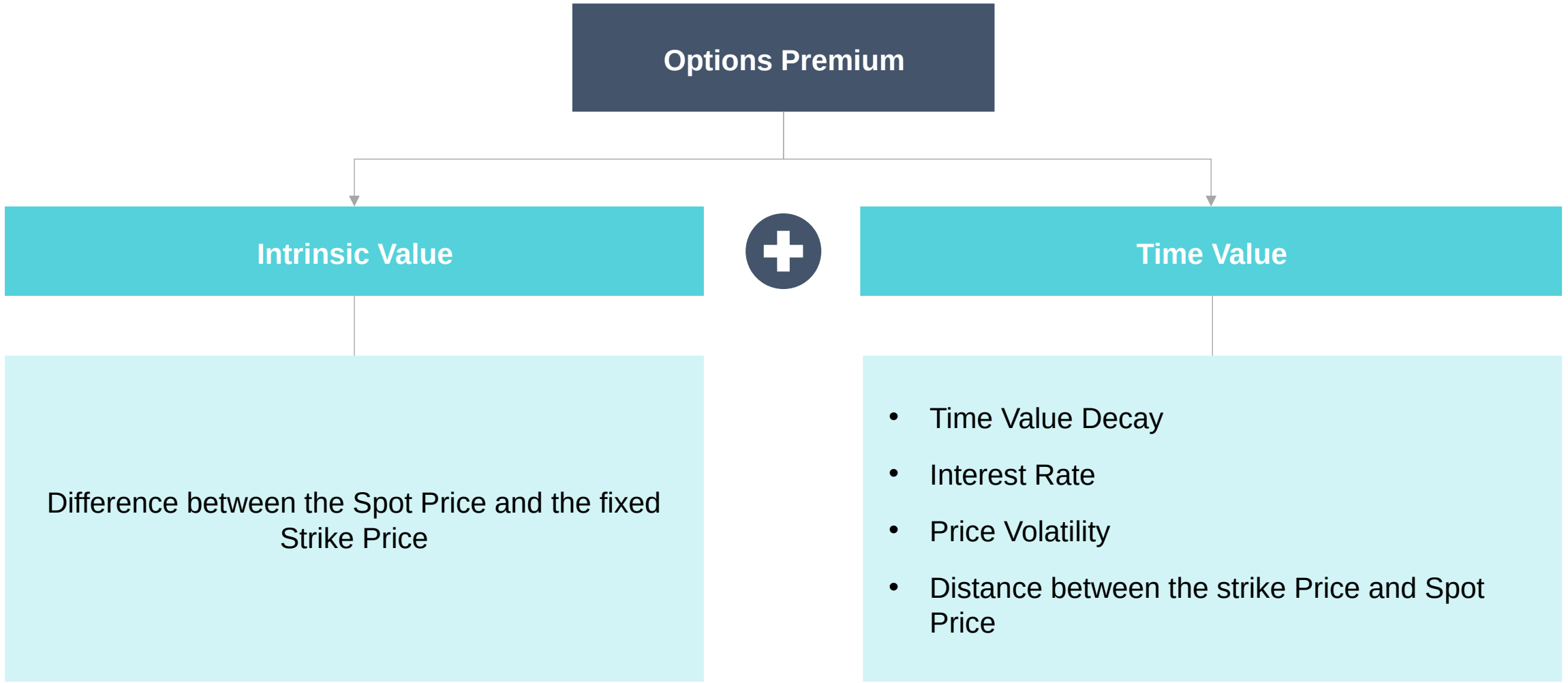
5. Intrinsic Value and Time Value

- **Intrinsic Value:** This is the portion of the premium that represents the immediate, realizable profit that an option would provide if it were exercised immediately. For call options, the intrinsic value is the difference between the current market price of the underlying asset and the strike price (if positive). For put options, it is the difference between the strike price and the market price of the underlying asset (if positive).
- **Time Value (Extrinsic Value):** The time value is the part of the premium that exceeds the intrinsic value. It reflects the potential for the option to gain additional value before expiration. Time value is influenced by factors like time to expiration, implied volatility, and the level of interest rates. As time passes, all else being equal, the time value of an option tends to decrease, a phenomenon known as time decay.

6. Factors Affecting Premium

- **Underlying Asset Price**: As the market price of the underlying asset moves relative to the option's strike price, the premium changes.
- **Time to Expiration**: Options with longer time to expiration typically have higher premiums because they have more time for potential price movements.
- **Volatility**: Higher implied volatility tends to increase the premium because it suggests a greater potential for price swings in the underlying asset.
- **Interest Rates**: Changes in interest rates can impact the premium, especially for options with longer time horizons.
- **Dividends**: For options on stocks, the timing and amount of dividends can affect premiums, particularly for call options.

OPTIONS PREMIUMS



FACTORS AFFECTING PREMIUMS







Time to Expiration	Stock Price and Strike	Volatility
American Options can exercise their right anytime till the expiration date, while European options can only exercise on a particular date. Higher the time duration, higher the premium.	The In-the-moneyness and Out of-the- moneyness will change depending on whether the Strike is higher than the Spot (Intrinsic Value). Therefore, this will determine if the option premium is higher or lower.	It measures how the price of a stock moves. Higher the volatility higher the premium as the chances of earning profits is higher and vice-versa.

FACTORS AFFECTING PREMIUMS





Risk Free Interest Rate

Generally speaking, there is a minimal impact on the option premium.

When the interest rates in the market go up, the returns on options as well are expected to go up, which in turn means a higher premium.

Dividends Expected

Generally speaking, the price of a stock reduces on an ex-dividend date.

If the option expiration falls on an ex-date or after Ex-Date, then the premiums will be lower if it's a call option, and higher if it's a put options.

TRADE ECONOMICS OF OPTIONS

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Economics	Meaning
Trade Date	The date on which the Option is agreed between the two parties
Settlement Date	The date on which the cash/securities of the option is moved between the two parties .
Exercise/Expiry Date	The dates on which the holder will decide if they will exercise or expire the option.
Premium	Premium amount which is paid by the holder to the writer.
Premium Payment Date	The date on which the premium is paid.
Exercise Type	American / European / Bermudan
Direction	Buy or Sell the option
Call / Put	Type of Options
Notional/Quantity	The Notional/Quantity of the underlying of the option
Strike	The price at which the holder will call or put the underlier.
Counterparty	The other party of the transaction (incase of OTC options).

TRADE ECONOMICS OF OPTIONS

Trade Economics of Options	
Trade Date:	
Option Style:	European /American/Bermudan
Option Type:	Call / Put
Seller:	Party A / Party B
Buyer:	Party A / Party B
Underlying Asset - Stock / Index:	
Number of Options:	
Strike Price:	
Premium:	
Premium per Option:	

Procedures for Exercise	
Expiration Time:	Valuation Time
Expiration Date:	Valuation Date
Multiple Exercise:	Not Applicable
Automatic Exercise:	Applicable

TRADE ECONOMICS OF OPTIONS

Valuation	
Valuation Time, Valuation Date:	As agreed in the Contract
Futures Price Valuation:	Applicable

Settlement Terms	
Cash Settlement:	Applicable
Settlement Currency:	Example : Japanese Yen
Cash Settlement Payment Date:	

OPTIONS TRADING



OPTION STRATEGIES - LONG CALL (BULLISH)

Let's remember the rules of Long Call: Long Call (Buy)



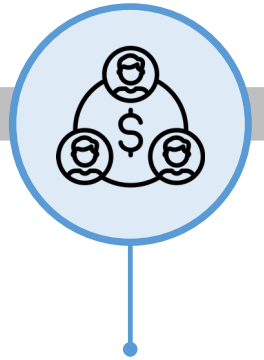
1. Buy Call Option Contract



2. Pay a Premium



3. Right to BUY the Underlying Asset



4. Holder of the Contract

Let's look at the scenarios of how a long call option becomes profitable:

- The buyer of a call option speculates that the price of the underlying asset goes up.
- If the price of the underlying asset goes up, i.e. the market price of the underlying asset is greater than the strike price then the option contract becomes profitable.
- If the price of the underlying asset goes down or remains the same, then the option contract becomes a loss.
- Option Contract Pay Off : Limited Loss (Max Premium paid) & Unlimited Profits Potential

OPTION STRATEGIES - LONG CALL EXAMPLE - Buying Call Option

Call Option Contract Details:

Underlying Asset: CISCO Systems Common Share

Strike Price: $50 per share

Premium Paid: $10 per share

Expiration Date: 30th Dec, 202X

Market Price	Strike Price	Premium	Exercise/Expiry	Profit and Loss
$30.00	$50.00	$10.00	Expire	$10.00 (-)
$40.00	$50.00	$10.00	Expire	$10.00 (-)
$50.00	$50.00	$10.00	Expire	$10.00 (-)
$60.00	$50.00	$10.00	Exercise	$0.00 (BEP)
$70.00	$50.00	$10.00	Exercise	$10.00 (+)
$80.00	$50.00	$10.00	Exercise	$ 20.00 (+)

Point to note: as the market price goes up the profit goes up and it's unlimited (theoretically).
The loss is limited to the premium paid.
if we Plot the above table on a Line Graph.

OPTION STRATEGIES - LONG CALL



OPTION STRATEGIES - SHORT CALL (BEARISH)

Let's remember the rules of Short Call: Short Call (Sell)



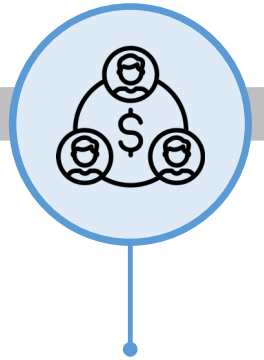
1. Sell Call Option Contract



2. Receive a Premium



3. Obligation to SELL the Underlying Asset



4. Writer of the Contract

Let's look at the scenarios of how a short call option becomes profitable:

- The seller of a call option speculates that the price of the underlying asset goes down.
- If the price of the underlying asset goes down, i.e. the market price of the underlying asset is less than the strike price then the option contract becomes profitable to the Seller of the Option.
- If the price of the underlying asset goes up or remains the same, then the option contract becomes a loss.
- Option Pay Off - Limited Profits (Max to the Premium Received) & Unlimited Loss Potential

OPTION STRATEGIES - SHORT CALL - Selling/Writing Call Options

Underlying Asset: CISCO Systems Common Share

Strike Price: $50 per share

Premium Paid: $10 per share

Expiration Date: 30th Dec, 202X

Market Price	Strike Price	Premium	Exercise/Expiry	Profit and Loss
$30.00	$50.00	$10.00	Expire	$10.00 (+)
$40.00	$50.00	$10.00	Expire	$10.00 (+)
$50.00	$50.00	$10.00	Expire	$10.00 (+)
$60.00	$50.00	$10.00	Exercise	$0.00 (BEP)
$70.00	$50.00	$10.00	Exercise	$10.00 (-)
$80.00	$50.00	$10.00	Exercise	$ 20.00 (-)

Point to note: as the market price goes up the losses goes up and it's unlimited loss (theoretically).
The Gain/Profit is limited to the premium received.
If we Plot the above table on a Line Graph.

Short Call Payoff



DIFFERENCE BETWEEN LONG CALL AND SHORT CALL

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Long Call

1. Option Buyer expects prices to go up
2. Premium is paid
3. When prices Go Up – there is Unlimited Profit
4. When prices Remain Same or Go Down - they lose premium paid

Short Call

1. Option Seller does not expect the prices to go up
2. Premium is received
3. When prices Go Up – there is Unlimited Loss
4. When prices Remain Same or Go Down - they Gain Premium received

OPTION STRATEGIES - LONG PUT (BEARISH)

Let's remember the rules of Long Put: Long Put (Buy)



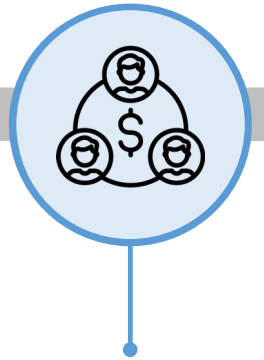
1. Buy Put Option Contract



2. Pay a Premium



3. Right to SELL the Underlying Asset



4. Holder of the Contract

Let's look at the scenarios of how a long-put option becomes profitable:

- The buyer of a put option speculates that the price of the underlying asset goes down.
- If the price of the underlying asset goes down, i.e. the market price of the underlying asset is less than the strike price then the option contract becomes profitable.

OPTION STRATEGIES - LONG PUT

Underlying Asset: CISCO Systems Common Share

Strike Price: $50 per share

Premium Paid: $10 per share

Expiration Date: 30th Dec, 202X

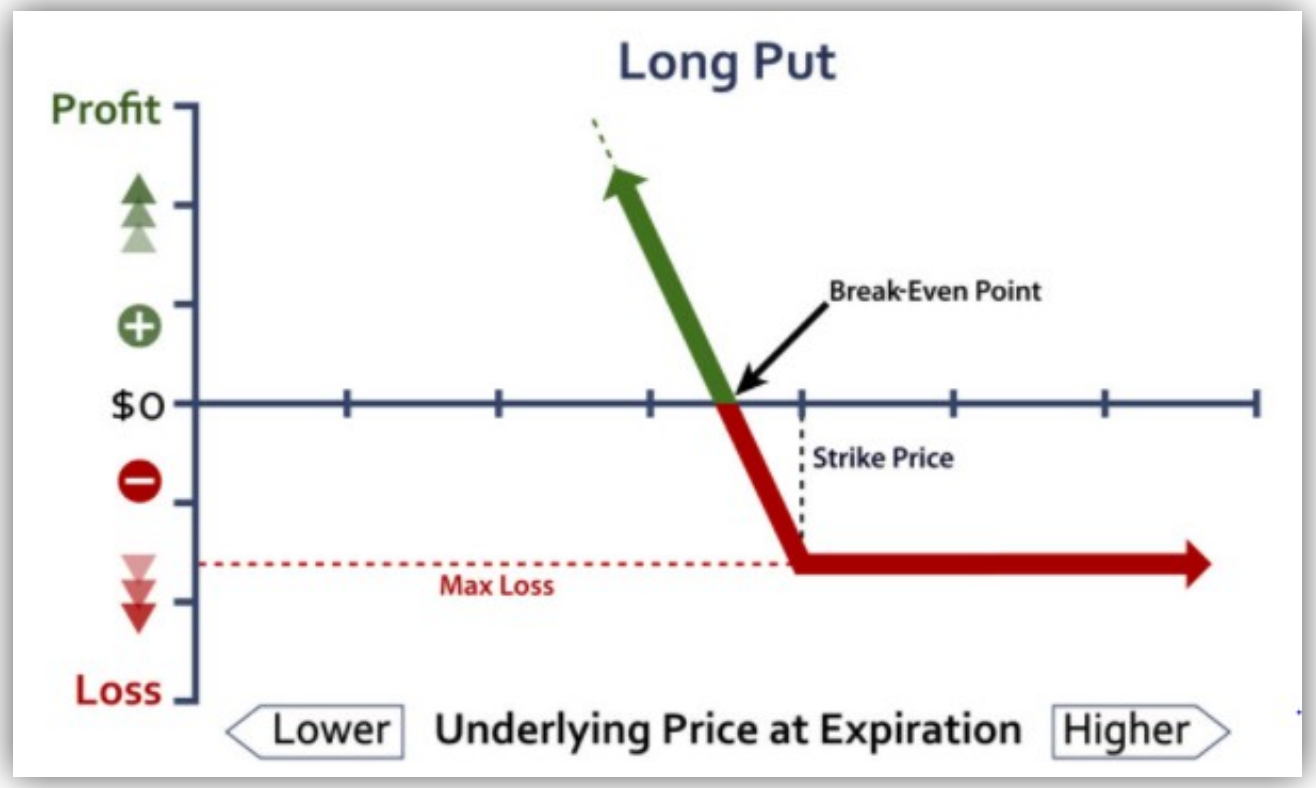
Market Price	Strike Price	Premium	Exercise/Expiry	Profit and Loss
$10.00	$50.00	$10.00	Exercise	$30.00 (+)
$20.00	$50.00	$10.00	Exercise	$20.00 (+)
$30.00	$50.00	$10.00	Exercise	$10.00 (+)
$40.00	$50.00	$10.00	Exercise	$0.00 (B.E.P)
$50.00	$50.00	$10.00	Expiry	$10.00 (-)
$60.00	$50.00	$10.00	Expiry	$10.00 (-)
$70.00	$50.00	$10.00	Expiry	$10.00 (-)
$80.00	$50.00	$10.00	Expiry	$10.00 (-)

Point to note: as the market price falls the profits up and it's unlimited profits (theoretically).
The Loss is limited to the premium paid.
If we Plot the above table on a Line Graph.

OPTION STRATEGIES

Point to note: as the market price goes down the profit goes up and it's limited as the stock price cannot go to zero. The loss is limited to the premium paid.

Long Put Payoff



OPTION STRATEGIES – SHORT PUT (BULLISH)

Let's remember the rules of Short Put: Short Put (Sell)



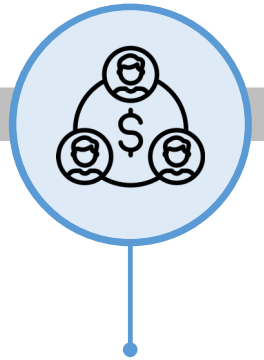
1. Buy Put Option Contract



2. Receives a Premium



3. Obligation to BUY the Underlying Asset



4. Writer of the Contract

Let's look at the scenarios of how a short put option becomes profitable:

- The seller of a put option speculates that the price of the underlying asset goes up.
- If the price of the underlying asset goes up, i.e. the market price of the underlying asset is more than the strike price then the option contract becomes profitable.
- If the price of the underlying asset goes down or remains the same, then the option contract becomes a loss.

OPTION STRATEGIES - SHORT PUT

Underlying Asset: CISCO Systems Common Share

Strike Price: $50 per share

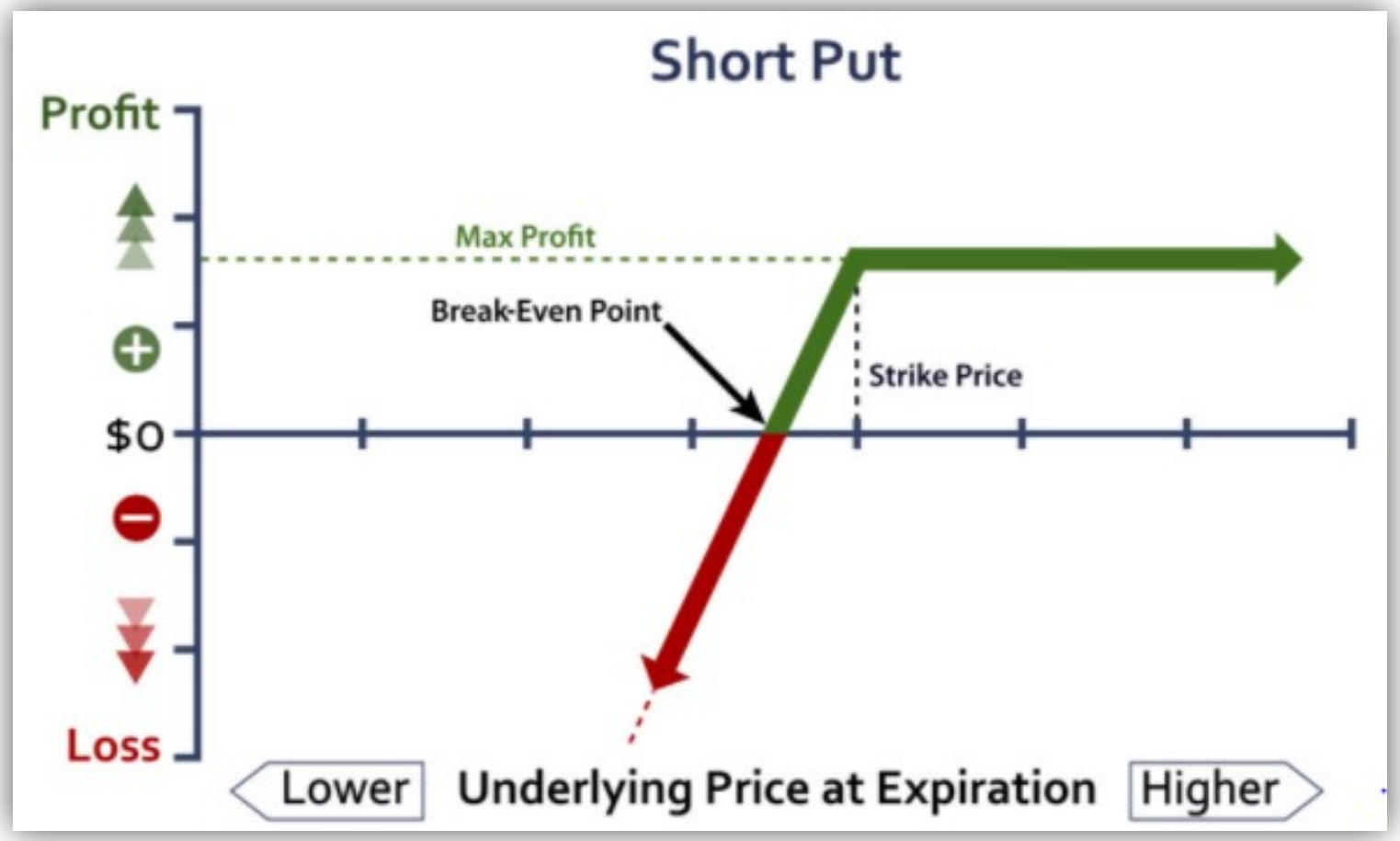
Premium Paid: $10 per share

Expiration Date: 30th Dec, 202X

Market Price	Strike Price	Premium	Exercise/Expiry	Profit and Loss
$10.00	$50.00	$10.00	Exercise	$30.00 (-)
$20.00	$50.00	$10.00	Exercise	$20.00 (-)
$30.00	$50.00	$10.00	Exercise	$10.00 (-)
$40.00	$50.00	$10.00	Exercise	$0.00 (B.E.P)
$50.00	$50.00	$10.00	Expiry	$10.00 (+)
$60.00	$50.00	$10.00	Expiry	$10.00 (+)
$70.00	$50.00	$10.00	Expiry	$10.00 (+)
$80.00	$50.00	$10.00	Expiry	$10.00 (+)

Point to note: as the market price goes up the profit goes up but it is limited to the premium paid. The loss is unlimited as well as a stock price falls.
If we plot the above the table in a line graph

Short Put Payoff



DIFFERENCE BETWEEN LONG PUT AND SHORT PUT

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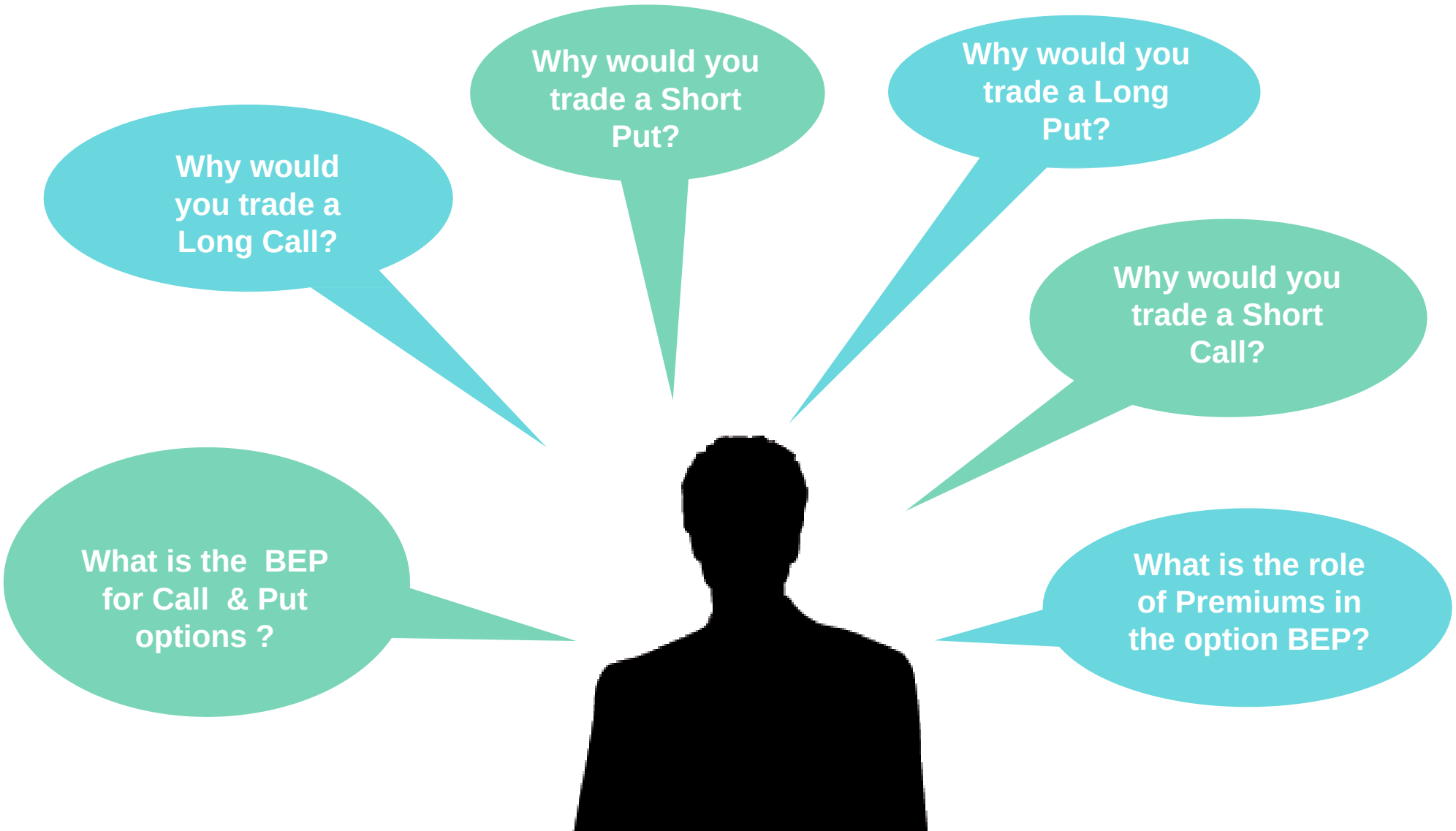
Long Put

1. Buyer expects prices to go down
2. Premium is paid
3. When prices Go Up – they Lose Premium
4. When prices Remain Same - they Lose Premium
5. When prices Go Down – there is limited Profit

Short Put

1. The buyer does not expect the prices to go down
2. Premium is received
3. When prices Go Up – they Gain Premium
4. When prices Remain Same - they Gain Premium
5. When prices Go Down – there is limited loss

QUESTIONS ON PAYOFFS



STATUS OF AN OPTION

The status refers to the pay off from exercising the option, and there are three statues as follows.

In the money (ITM)	Exercise will result in benefit to buyer. The Option Premium will be High.
At the money (ATM)	Exercise has neither benefit nor loss to buyer. The Option Premium will be Fair Value.
Out of the money (OTM)	Exercise will result in loss to the buyer. Hence the holder will expire the option. The Option Premium will be Cheaper.







OPTIONS PAYOFF

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Call Options Payoff

In the Money(ITM)

- CMP is greater than the Strike Price
- If the Call Option Buyer exercise the option, they will receive the cash flow

At the Money (ATM)

- CMP is equal to the Strike Price
- If the Call Option buyer exercise the option, they will not receive the cash flow

Out of the Money (OTM)

- CMP is lesser than the Strike Price
- Call Option Buyer dont exercise the contract and let the Option Lapse, they will not receive the cash flow

STATUS OF AN OPTION

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Call Options

Type	Strike Price	Market/Spot Price	Option
Call	90	100	ITM
Call	100	100	ATM
Call	110	100	OTM

Put Options Payoff

In the Money

- CMP is lesser than the Strike Price
- If the buyer exercise the option, they will receive the cash flow

At the Money

- CMP is equal to the Strike Price
- If the buyer exercise the option, they will not receive the cash flow

Out of the Money

- CMP is greater than the Strike Price
- Buyers dont exercie the option and let the Option Lapse, they will not receive the cash flow

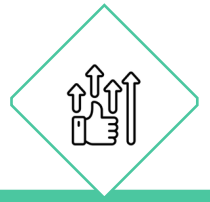
STATUS OF AN OPTION

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Put Options

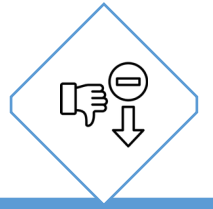
Type	Strike Price	Market/Spot Price	Option
Put	90	100	OTM
Put	100	100	ATM
Put	110	100	ITM

PROS AND CONS OF OPTIONS TRADING



Advantage

- Leverage
- Risk/reward ratio
- Unique strategies
- Low capital requirements
- Diversification



Disadvantage

- Lower liquidity
- Higher spreads
- Higher commissions
- Complicated
- Time decay
- Less information
- Options not available for all stocks

OPTION CHAIN

Option Chain is a way to quote option prices in a list with all the options for a given security. This listing has all the Call and Put Options along with the premium payable and their maturity.

RELIANCE 30Sep2025 CE 1300.00

NSE LIVE **83.00** ▲ **9.90** (13.54%)
Sep 05, 15:19

BSE: 500325 | NSE: RELIANCE | ISIN: INE002A01018

Open Price	76.95	Average Price	77.56
High Price	87.50	No. of Contracts Traded	379
Low Price	69.60	Turnover (Rs. in lakhs)	2,610.48
Prev. Close	73.10	Market Lot	500
Spot Price	1,375.00	Open Interest	691,000
Open Int PCR	0.64	Open Int. Chg	-50,000
Prev OI PCR	0.62	Open Int. Chg %	-6.75
Bid Price	82.30	Offer Price	82.85
Bid Qty	2,000	Offer Qty	500

Option Chain Snapshot

NEAR | NEXT | FAR — Enter Strike Price Range: From — To — GO

Calls					Strike Price	Puts				
LTP (Rs)	Net Change	Volume	Open Interest	Change In Open Int		LTP (Rs)	Net Change	Volume	Open Interest	Change In Open Int
49.10	6.75	1,023,500	330,500	-113,000	1,340.00	10.40	-5.35	2,819,500	1,331,000	-113,000
42.00	5.85	1,996,000	1,110,000	-243,000	1,350.00	13.30	-6.40	3,316,500	2,137,500	-146,500
35.80	5.20	5,479,500	2,436,000	-768,500	1,360.00	16.85	-7.10	5,079,000	2,860,500	-201,500
29.95	4.30	7,716,000	2,381,000	-922,500	1,370.00	21.00	-7.95	3,953,000	1,978,000	-105,500
24.90	3.65	7,607,000	4,431,500	-614,000	1,380.00	25.95	-8.65	3,046,500	3,013,500	-143,000
20.50	3.15	3,108,000	2,544,000	-230,500	1,390.00	31.30	-9.50	934,500	1,839,000	-24,000
16.70	2.45	10,325,000	12,230,500	-481,500	1,400.00	37.75	-9.90	1,275,000	5,988,000	73,500



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THANK YOU !!!

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